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Original article

# Safety and efficacy profile of the fluocinolone acetonide implant in non-infectious uveitic macular edema: 5-year follow-up

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ABSTRACT

*Objective:* To evaluate the safety and efficacy profile of the fluocinolone acetonide implant (FAI) in patients with non-infectious posterior segment uveitis over a 60-month follow-up period.

*Methods:* We retrospective conducted a study with a mean follow-up of 47 months (minimum, 18; maximum, 60). The study included 11 eyes (9 patients), with a male-to-female ratio of 8:1. The etiology of uveitis was heterogeneous. Efficacy parameters included visual acuity (VA) and Central macular thickness (CMT), and the ones who needed adjuvant therapy or re-injection. Safety parameters focused on the development of cataracts and intraocular hypertension.

*Results:* Real-world outcomes were assessed. Both VA recovery and CMT reduction were achieved between 3 to 6 months after FAI implantation and remained stable at the follow-up. All patients but one were pseudophakic prior to implantation. The phakic patient developed early cataract formation, requiring surgery 9 months post-implantation. All cases had previously received intravitreal dexamethasone. Five eyes (45.4%) required topical hypotensive therapy, and none glaucoma surgery. No patient required periodic adjuvant therapy or additional FAI injections, even among those with the longest follow-up periods. Two cases (18.1%) exhibited mild inflammatory activity in the macular area at months 33 and 35 post-implantation. These episodes were successfully resolved with sub-Tenon triamcinolone injections, maintaining clinical stability thereafter.

*Conclusions:* The fluocinolone acetonide implant (FAI), in our series of cases, turned out to be an effective and safe therapy for non-infectious uveitic macular edema, providing sustained and long-lasting effects, in some cases up to 60 months post-implantation. Sub-Tenon triamcinolone proves to be a valuable adjunctive therapy.



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## Eficacia y seguridad del implante de acetónido de fluocinolona en edema macular y uveítis posterior no infecciosa: seguimiento de 5 años

*Palabras clave:*
Implante acetónido fluocinolona
Edema macular uveítico
Uveítis posterior no infecciosa

R E S U M E N

*Objetivo:* Valorar la eficacia y seguridad del implante de acetónido de fluocinolona (iAF) en pacientes con uveítis posterior no infecciosa durante un periodo de seguimiento de hasta 60 meses.

*Método:* Estudio retrospectivo, seguimiento medio 47 meses (mín 18 y máx 60). 11 ojos (9 pacientes), 8:1 (M:V). La etiología es heterogénea. Los parámetros de eficacia son agudeza visual (AV) y grosor macular central (GMC), así como si precisan tratamiento adyuvante o reinyección. Los parámetros de seguridad son el desarrollo de catarata e hipertensión ocular.

*Resultados:* Resultados de vida real. Tanto la recuperación de AV como la reducción del GMC se alcanza entre los 3-6 meses tras iAF y se mantienen estables a lo largo del seguimiento. Todos los casos salvo uno eran pseudofáquicos en el momento del implante. El paciente fáquico desarrolló catarata precoz, se intervino 9 meses tras el implante. Todos los casos habían llevado previamente dexametasona intravítrea. 5(45,4%) requieren tratamiento hipotensor tópico y ninguno precisó cirugía de glaucoma. Ninguno los casos precisó tratamiento adyuvante periódico ni tampoco una reinyección iAF, incluidos los pacientes con mayor periodo de seguimiento. Dos de los casos (18,1%) presentaron una discreta actividad inflamatoria en área macular en los meses 33 y 35 tras iAF y se resolvió con una inyección de triamcinolona subtenoniana, manteniéndose la estabilidad clínica.

*Conclusiones:* El iAF ha resultado ser un tratamiento eficaz y seguro en nuestra serie en edema macular uveítico no infeccioso con un efecto mantenido y prolongado en el tiempo, en algunos casos hasta 60 meses tras el implante. La triamcinolona subtenoniana es útil como tratamiento adyuvante.



## Introduction

Uveitis is a group of diseases characterised by inflammation affecting the intermediate layer of the eye, the uvea. Prevalence ranges between 58 and 121 cases per 100,000 inhabitants,[1] varies according to race and geographic location, and is more frequent in women because of their higher prevalence of rheumatic and autoimmune diseases. Several classifications exist; based on the ocular structures involved, uveitis is defined as anterior, intermediate, posterior, or panuveitis.[2]

Macular edema is the leading cause of decreased visual acuity (VA) in uveitis.[3] It is present in 40%–45% of cases and is produced by the action of inflammatory mediators on the blood-retinal barrier.[4] It is more common in posterior uveitis. When inflammation becomes chronic or highly recurrent, functional and anatomical alterations occur in the retina, irreversibly damaging the tissues.[4]

More than 50% of uveitis cases are of unknown etiology.[5] When an identifiable cause exists, they are classified as infectious or immune. Posterior uveitis in which an infectious cause is excluded are collectively referred to as noninfectious posterior uveitis (NIPU).[6] This group encompasses very heterogeneous entities: autoimmune diseases with isolated ocular involvement (eg, Fuchs iridocyclitis, multifocal choroiditis, Birdshot chorioretinopathy) and those associated with systemic disorders (eg, Behçet disease, Vogt-Koyanagi-Harada disease, ankylosing spondylitis, sarcoidosis, collagenopathies, or juvenile rheumatoid arthritis).[7]

The goal of NIPU treatment is to control inflammatory activity and prevent recurrences. Each recurrence produces retinal structural damage that, although sometimes anatomically reversible, leaves cumulative injury that impairs function.[8]

Treatment intensity depends on whether systemic disease or isolated ocular involvement is present, as well as on the patient's risk profile and risk of adverse effects.[7]

Treatment options include corticosteroids (systemic, topical, periocular, intravitreal), antimetabolites, calcineurin inhibitors, alkylating agents, and biologic therapies.[8] In treatment algorithms for NIPU, corticosteroids—local or systemic—are first-line therapy, followed by immunosuppressive therapies.[10]

Corticoid implants are a good alternative for patients who do not respond to or cannot tolerate immunomodulatory therapy and are especially useful for cystoid macular edema.[7] Systemic treatments, whether corticosteroids or immunomodulators, are associated with significant adverse effects; therefore, priority has been given to developing

**Table 1 – Demographic Characteristics of the Sample and Etiology of Posterior Uveitis.**

| Case | Sex | Laterality | Age (y) | Etiology | Systemic treatment |
|---|---|---|---|---|---|
| 1 | Female | OS | 88 | Artisan IOL, HLA-B27+ | – |
| 2 | Female | OU | 77 | Pars planitis, ANA+ | MTX (discontinued) |
| 3 | Female | OD | 45 | Adolescent uveitis, CMV+ | – |
| 4 | Female | OD | 58 | Arthropathy, HLA-B27+ | – |
| 5 | Female | OS | 51 | Birdshot disease | MTX |
| 6 | Female | OU | 75 | Posterior uveitis, HCV+ | – |
| 7 | Male | OD | 63 | Vasculitis | MTX |
| 8 | Female | OD | 69 | Behçet disease | – |
| 9 | Female | OS | 73 | Pars planitis | – |

OD, right eye; OS, left eye; OU, both eyes; LIO, intraocular lens; ANA, antinuclear antibody; CMV, cytomegalovirus; HCV, hepatitis C virus; MTX, methotrexate.

drugs implanted in the vitreous cavity, which act locally and avoid systemic side effects.[9] Compared with dexamethasone, fluocinolone implants offer the advantage of longer duration, avoiding reinjections every few months. Fluctuations in inflammatory activity with anatomical and functional impact increase the risk of permanent visual impairment.[9]

The fluocinolone acetonide implant (FAi) is a synthetic fluorinated glucocorticoid. Initially approved for diabetic macular edema, it has since been approved for the treatment of NIPU and for prevention of recurrences.[10] The non-biodegradable device releases 0.2 μg/day. No systemic drug has been detected after intraocular administration.[7] The FAMOUS trial demonstrated sustained drug release and stable vitreous concentrations for 36 months.[11]

Use is contraindicated in established glaucoma, optic cup-to-disc ratio >0.8, and suspected intra- or periocular infection.[12] It is also contraindicated in patients with hypersensitivity to implant components such as adhesive silicone, polyvinyl alcohol, or polyamide tubing.[12]

FAi acts by inhibiting inflammatory mediators in the arachidonic acid cascade and reducing intravitreal VEGF levels.[12]

A European expert group developed a treatment algorithm for diabetic macular edema using FAi, evaluating both monotherapy and adjunctive use.[9] It is critical to assess whether inflammatory activity reflects a purely ocular disease or a sign of systemic disease, and whether involvement is unilateral or bilateral.

Few studies have evaluated the long-term efficacy and safety of FAi in NIPU; those available often report follow-up shorter than the drug's half-life. A European phase 3 prospective, double-masked, multicenter trial demonstrated favourable efficacy and safety results over 36 months, including patients with macular edema of various etiologies.[13]

## Methods

We conducted a retrospective study of NIPU cases treated with FAi 0.19 mg (ILUVIEN®, Alimera Sciences, Alpharetta, GA) between July 2019 and April 2023 at *Hospital Clínico Universitario de Valencia* (Valencia, Spain). The study was approved by the institutional research ethics committee and adhered to the principles set forth in the Declaration of Helsinki. All patients gave oral and written informed consent before treatment.

Inclusion criteria were receipt of FAi during the study period as treatment for macular edema secondary to NIPU. Data were collected from electronic health records. Exclusion criteria were uveitis of infectious origin, uncontrolled ocular hypertension or glaucoma, and pregnancy or breastfeeding.

All patients were adults with posterior uveitis, pars planitis, with or without anterior uveitis, and all had macular edema.

The mean follow-up period was 47 months (range, 18–60). All patients had previously received at least one dexamethasone implant before FAi; 7 cases (63%) had also received intravitreal anti-VEGF injections. After FAi, patients were reviewed quarterly within the first 2 years and then every 6 months, according to clinical status.

Clinical evaluation criteria included VA using the decimal system, intraocular pressure (IOP) measured with Goldmann applanation tonometry, wide-field fundus photography (Clarus 700, Carl Zeiss, Jena, Germany), and central macular thickness (CMT) using swept-source OCT (Triton OCT, Topcon Corporation, Tokyo, Japan).

Measurements were obtained before FAi, quarterly for the first 2 years, and every 6 months thereafter.

Statistical analysis was performed using SPSS version 16 (IBM Corp).

## Results

The cohort included 11 eyes from 9 patients: 4 right eyes, 3 left eyes, and 2 patients with bilateral implants. Mean age was 66.6 years (SD, 13.6; range, 45–88). The male-to-female ratio was 1:8. All but 1 patient were pseudophakic at the time of FAi injection.

All cases presented NIPU-induced macular edema, though etiologies were heterogeneous (Table 1). Three cases (27.3%) had undergone vitrectomy (2 prior to FAi and 1 after, due to lens-sack dislocation). The latter was a patient with posterior

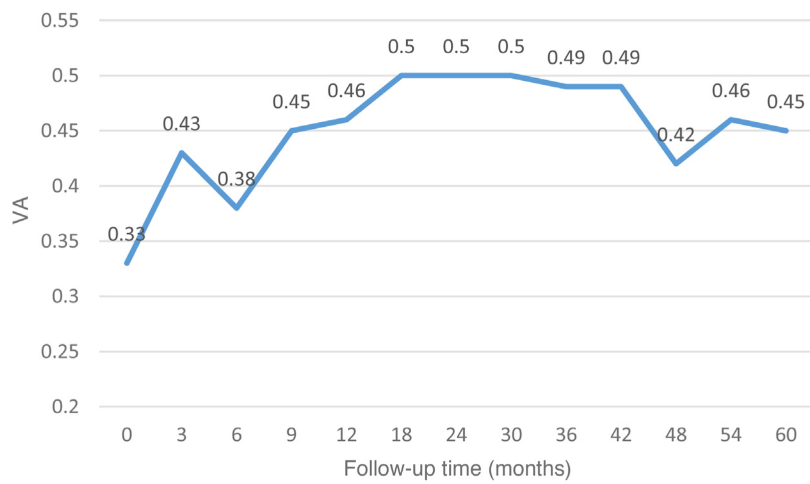


**Figure 1 – Evolution of mean visual acuity (VA) at the 60-month follow-up. The decline in VA at month 6 corresponded to the development of cataract in the only phakic case, subsequently restored after cataract surgery. The follow-up period was not equal across the cohort.**

uveitis beginning in adolescence and cataract surgery at age 22 after chronic corticosteroid use.

The mean follow-up period was 47 months (range, 18–63). Two patients were lost to follow-up: 1 after 18 months, and 1 after implantation in early 2020 during the COVID-19 pandemic, who returned after 36 months. Data were collected from the time of return.

The efficacy of FAi was evaluated by 2 clinical parameters: VA and CMT. Fig. 1 illustrates the evolution of mean VA; the baseline value corresponded to the last measurement before implantation. A limitation of the curve is that not all patients had the same follow-up time, and the further to the right the curve extends, the fewer eyes are included in the mean; specifically, after 48 months the values represent the mean of 3 eyes. Nevertheless, mean VA before injection was 0.33 and increased in all cases, remaining stable, though not constant, through 60 months of follow-up. The notch in the curve at 6 months, subsequently corrected, corresponded to the development of a dense cataract in one patient, which affected the mean value; the mean improved again after cataract surgery. Comparison of mean VA before implantation and at last follow-up, using the Wilcoxon signed-rank test for paired samples, showed a statistically significant improvement in VA ($p = .02$).

The anatomical effect of FAi was reflected in the progression of CMT, represented in Fig. 2. Normalization of mean CMT was achieved within the first 6 months and remained constant throughout follow-up. The anatomical effect was more stable than the functional effect, attributable to the potent and long-lasting anti-inflammatory activity of the drug. However, for optimal functional recovery, preservation of the outer retinal layers is also required, which is not always the case in chronic disease. Comparison of baseline and final CMT values for each patient, accounting for differing follow-up times, showed a statistically significant reduction by the Wilcoxon signed-rank test for paired samples ($p = .008$).

Two adverse events were observed in this series: cataract and ocular hypertension. All but 1 patient were pseudophakic at the time of injection. The single phakic patient developed an early cataract and underwent surgery 9 months after injection.

All eyes had received prior dexamethasone implantation. As a preventive measure, all cases were managed with topical IOP-lowering therapy, which was maintained in patients whose IOP remained at the high end of the normal range. Five cases (45.5%) were already on topical hypotensive therapy before FAi, and this percentage remained unchanged. In one case, treatment was escalated from 1 to 2 agents to normalize IOP values. The course of mean IOP values is shown in Fig. 3. No eyes required explantation of the device or glaucoma surgery. Patients on topical therapy before FAi were the same ones requiring therapy afterward. The IOP curve was slightly upward but always within normal limits. Comparison of baseline and final IOP by the Wilcoxon signed-rank test revealed no statistically significant differences ($p = .89$).

No cases required retreatment, and no FAi reimplantations were performed, even in patients followed for 60 months. In 2 cases with mild recurrent macular edema, adjunctive sub-Tenon triamcinolone injection was administered prior to FAi reimplantation, resulting in edema resolution and subsequent stability. Triamcinolone rescue therapy was performed at months 33 and 36 of follow-up, respectively.

Three patients were on systemic methotrexate; in one of these (Case report #2), systemic therapy was discontinued and disease was controlled with FAi alone. This was one of the bilateral cases with the longest follow-up in the series.

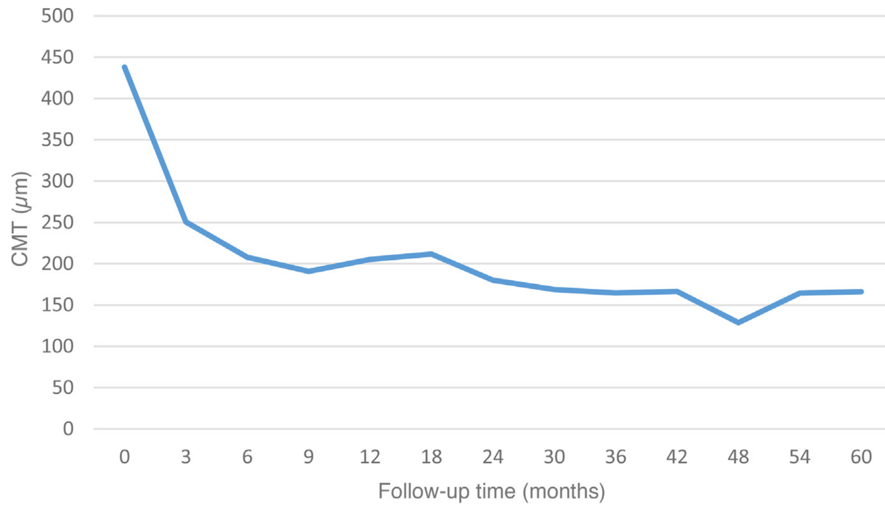


**Figure 2 – Evolution of mean CMT. Mean CMT normalized within the first 6 months and remained stable throughout follow-up.**

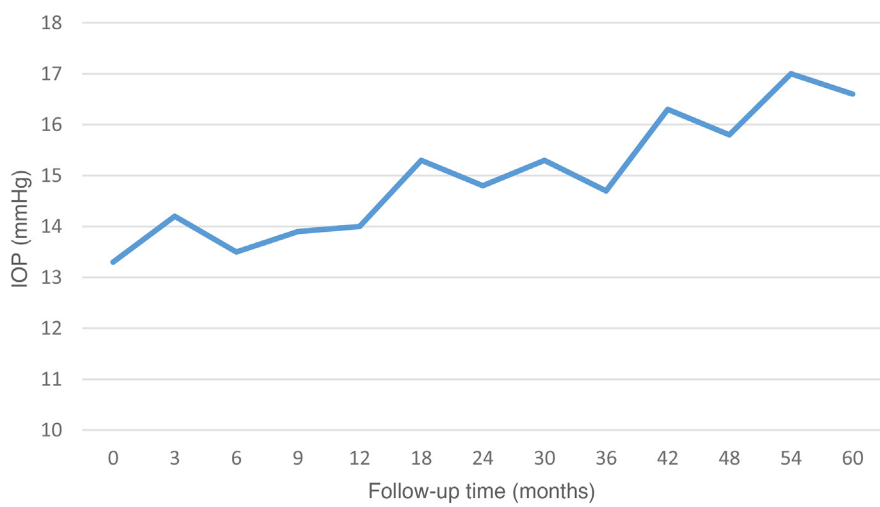


**Figure 3 – Evolution of mean IOP after FAi.**

## Discussion

FAi proved effective and safe in controlling NIPU-induced macular edema in this cohort. The sample was heterogeneous in both clinical presentation and etiology, limiting generalisability. Several published meta-analyses have reported safety and efficacy outcomes of FAi.[12,14,15]

If cases are classified according to whether the involvement is purely ocular or a sign of systemic disease, treatment with FAi may be effective for a long period without the need for retreatment. In patients with exclusively ocular, bilateral signs, systemic therapy and its adverse effects may even be avoided. Long-term visual prognosis depends largely on the patient's baseline condition, disease duration, the number and aggressiveness of relapses, and baseline retinal anatomy. In this regard, it is important to evaluate biomarkers at the time of indication.[7] Some of these OCT biomarkers include the presence of hyperreflective spots, neurosensory detachment, intraretinal cysts, and disruption of the inner and/or outer retina. There is a direct relationship between functional improvement and retinal structure.

The drug is effective for a long period, both for treatment and prevention of recurrences. Once the indication has been established, initiation of treatment should not be delayed.

All patients in our series had previously received a dexamethasone implant. Some studies suggest induction therapy prior to FAi with dexamethasone implant or sub-Tenon triamcinolone injection.[7] Some authors have proposed a provocation test before implantation as a useful measure to rule out corticosteroid responders and improve safety in treated patients.[9] In most reviews, the rate of IOP-lowering treatment in uveitis is approximately 31% and glaucoma surgery 3.9%.[7] In our series, the rate of patients treated (45%) was slightly higher, as many were already receiving topical therapy before implantation; however, the glaucoma surgery rate was lower. We did not find patients with IOP > 30 mmHg, as reported in other reviews.[4] Nor did we observe adverse events described elsewhere, such as hypotony, vitreous hemorrhage, or anterior migration of the implant.[7] The FAME study[16] focused on drug safety, regardless of underlying disease, reporting that approximately 13% of patients developed IOP > 31 mmHg, 26% required topical treatment, and 14% required glaucoma surgery.[17] It is possible that glaucoma development in uveitis is less common than in diabetes, given that diabetes is a multifactorial disease in which not only inflammation but also associated vasculopathy contributes to ocular hypertension and/or glaucoma.

The risk of cataract development varies across studies. In the FAME trial, with 36 months of follow-up, cataract developed in 81.7% of cases,[17] consistent with our results.

The main limitation of this study is the small sample size and heterogeneity in uveitis etiology.

In our cohort, FAi proved to be an effective treatment for NIPU, maintaining stable VA and CMT without fluctuations, in some cases up to 60 months after injection. To our knowledge, no published results report such long follow-up. FAi was effective both for treatment and prevention of recurrences, protecting against cumulative external retinal damage that irreversibly limits long-term VA. Rescue therapy with sub-Tenon triamcinolone injection is a simple and effective technique that may be used to control inflammatory recurrences of the posterior pole.

## Informed consent

All patients signed informed consent for the procedure and for use of images obtained during follow-up.

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None declared.

## Declaration of competing interest

None declared.

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